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Form factors in semileptonic decay $D \to Kl\bar{\nu}$ with Lattice QCD

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Form factors in decay $D \to Kl\bar{\nu}$ are researched by using Lattice QCD with 2+1-flavor domain wall fermion configurations. Valance quark action were chosen as Overlap fermion. The lattice size is $32^3 \times 64$ with a 2.383(9) GeV inverse lattice spacing. Form factors form pseudoscalar meson to pseudoscalar meson defined by both the scalar, vector and tensor currents are computed with a renormalization $\overline{MS}$ scheme at 2 GeV.

① Motivation

Form factors are important nonperturbative quantities in researching meson weak decays, and they can help determine the Cabibbo-Kobayashi-Maskawa (CKM) matrix elements which are the fundamental parameters of standard model (SM).

For higher precision of those parameters, not only more precise experimental measurements are required, but also precise calculations of Lattice QCD are required.

V_{cs} in CKM matrix can be extracted form the semileptonic decay $D \to Kl\bar{\nu}$, whose accuracy plays an indispensable role in searching of new physics.

② formulas and Lattice Setup

The form factors f_0, f_+ and f_T are defined as,

$$\langle K(p')|\bar{s}c|D(p)\rangle = \frac{m_D^2 - m_K^2}{m_c - m_s} \boldsymbol{f_0(q^2)},$$

$$\langle K(p')|\bar{s}\gamma_\mu c|D(p)\rangle = \boldsymbol{f_+(q^2)}\left[p_\mu + p'_\mu - \frac{m_D^2 - m_K^2}{q^2} q_\mu\right] + \boldsymbol{f_0(q^2)} \frac{m_D^2 - m_K^2}{q^2} q_\mu,$$

$$\langle K(p')|\bar{s}\sigma_{\mu\nu} c|D(p)\rangle = \frac{2}{m_D + m_K}[p_\mu p'_\nu - p_\mu p'_\nu] \boldsymbol{f_T(q^2)}.$$

Those hardon matrix element can be extracted from correlation functions calculated by LQCD.

In this work, the gauge configurations were generated by the RBC/UKQCD collaborations as table.1 show [1] . And valance quark is overlap fermion and numerical values of quark mass are as table.2 shows.

[1] RBC and UKQCD Collaborations, Phys. Rev. D 93(2016) 074505

am_l/am_s	m_π(MeV)	$N_{\text{conf}} \times N_{\text{src}}$
0.004/0.03	302	628×1
0.006/0.03	360	42×16
0.008/0.03	412	49×16

Table.1 The configuration ensembles. As the ratio of dynamical light quark mass to strange quark mass increasing, those ensemble are labeled as f004, f006 and f008. f006 was used for this preliminary result.

$am_l^{(\text{val})}$	0.00460, 0.00585, 0.00677, 0.00765, 0.00885 0.01120, 0.01290, 0.01520, 0.01800, 0.02400
m_π	220MeV~500MeV
$am_s^{(\text{val})}$	0.037, 0.040, 0.043, 0.046, 0.049, 0.052
$am_c^{(\text{val})}$	0.450, 0.492, 0.500, 0.550

Table.2 Valance quark masses. The maximum light quark mass and all strange quark masses and all charm quark masses are used for this preliminary result.

③ Numerical Analyses

Two point correlation functions of still D meson and K meson with multiple momentum and corresponding three point correlation functions form D to K were calculated to extract form factors. And a jointly least chi2 fitting method are used to get those hardon matrix elements and energies of meson with different momentum from correlation functions. The formalism of fitting functions are,

$$C^D(t,\vec{p}) = \frac{Z_D^2}{2E_D}\left[e^{-E_D t} + e^{-E_D(n_t - t)}\right],$$

$$C^K(t,\vec{p}') = \frac{Z_K^2}{2E_K}\left[e^{-E_K t} + e^{-E_K(n_t - t)}\right],$$

$$C^{D\to K}(t,\vec{p},\vec{p}') = \frac{Z_K}{2E_K}\frac{Z_D}{2E_D}\left[e^{-E_D t}e^{-E_K(T-t)} + e^{-E_D(n_t - t)}e^{-E_K(T-n_t+t)}\right].$$

④ Preliminary result

Form factor $f_0(q^2)$ and $f_+(q^2)$ can be determined form the hadron matrix elements for the temporal vector and scalar currents on each configuration. When K meson is moving, we also can add corresponding special vector currents for a system error which comes from different operators. And the form factor $f_T(q^2)$ can be extracted form the tensor currents which appears in theoretical model beyond SM.

An example of the results of $f_0(q^2)$ and $f_+(q^2)$ are as fig1. shows, and results of $f_T(q^2)$ are as fig.2 shows.

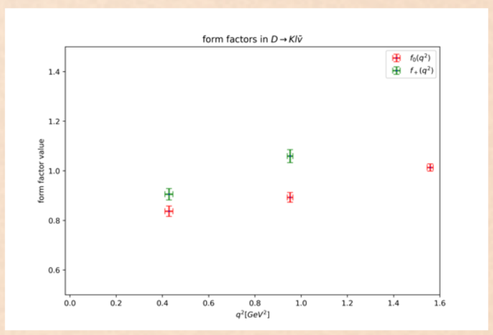


Fig.1 Form factor $f_0(q^2)$ and $f_+(q^2)$ at three different q^2 with light quark mass $am_l = 0.024$ and strange quark mass $am_s = 0.037$ and charm quark mass $am_c = 0.45$.

Values of Form factors are changing with the valance quark masses used in correlation functions, this relation has been studied, an example of form factor $f_0(q^2)$ is as fig.3 shows.

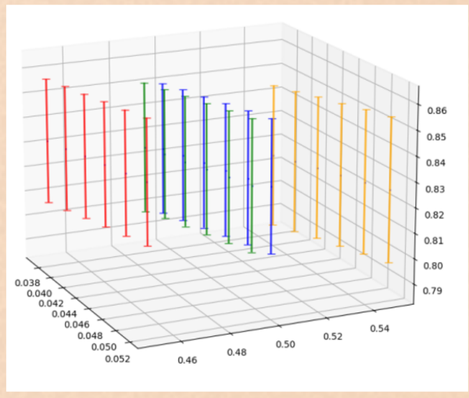


Fig 3. Behavior of form factor $f_0(q^2)$ as a function of valance quark masses when D meson is still and K meson has a momentum $ap_K = [0, 1, 1]$.

⑤ Outlook

Only the heaviest light quark mass was used, more data need to be calculated with more light quark masse.

With interpolate or extrapolate, valance quark masses can be